HANS-PAUL FREDERICK BAEHR

Current as of January 2026

BASIC INFORMATION

Citizenship: USA
Languages: native English, conversational German
Current position: Postdoctoral researcher; Max Planck Institute for Astronomy, Heidelberg
Current supervisor(s): Prof. Hubert Klahr

CONTACT

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Heidelberg, 69117
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EDUCATION

Doctor rerum naturalium (Ph.D.), magna cum laude, Astronomy – July 5, 2019
Thesis: Formation Criteria and Initial Constraints on Objects Formed in Self-Gravitating Disks
Ruprecht-Karls University, Heidelberg, Germany
Max Planck Institute for Astronomy, Heidelberg, Germany

Master of Science, Physics - March 20, 2015
Thesis: Conditions for Planet Formation via Gravitational Instability of Self-Gravitating Disks
Ruprecht-Karls University, Heidelberg, Germany
Max Planck Institute for Astronomy, Heidelberg, Germany

Bachelor of Science, Physics - May 4, 2012
Bachelor of Science, Mathematics - May 4, 2012
University of Utah, Salt Lake City, Utah, United States

EMPLOYMENT HISTORY

Postdoctoral Scholar: September 2024 – present
MPIA

Postdoctoral Scholar: September 2022 – August 2024
University of Georgia

Postdoctoral Scholar: September 2019 – August 2022
University of Nevada, Las Vegas

FIRST AUTHOR PUBLICATIONS

Baehr H., Klahr H., 2015, *The Role of the Cooling Prescription in Disk Fragmentation: Numerical Convergence & the Critical Cooling Parameter in Self-Gravitating Disks*, ApJ, 814, 155

Baehr H., Klahr H., Kratter K. M., 2017, *The Fragmentation Criteria in Local Vertically Stratified Self-Gravitating Disk Simulations*, ApJ, 848, 40

Baehr H., Klahr H., 2019, *The Concentration and Growth of Solids in Fragmenting Circumstellar Disks*, ApJ, 881, 162

Baehr H., Zhu Z., 2021, *Particle Dynamics in 3D Self-gravitating Disks I: Spirals*, ApJ, 909, 135

Baehr H., Zhu Z., 2021, *Particle Dynamics in 3D Self-gravitating Disks II: Dust Settling and Diffusion*, ApJ, 909, 136

Baehr H., Zhu Z., Yang C.-C., 2022, *Direct Formation of Planetary Embryos in Self-Gravitating Disks,* ApJ, 933, 100

Baehr H., 2023, *Filling in the Gaps: Can Gravitationally Unstable Discs Form the Seeds of Gas Giant Planets?,* MNRAS, 523, 3348

Baehr H., Rice K., Yang C.-C., Hall C., 2025, *On the Gravitational Collapse of Small Dust Grains in Self-gravitating Disk Structures,* ApJ, 995, 39

CONTRIBUTING AUTHOR PUBLICATIONS

Gibbs A., …, **Baehr H.**, et al., 2020, *EDEN: Sensitivity Analysis and Transiting Planetary Detection Limits for Nearby Late Red Dwarfs*, AJ, 159, 169

Gerbig K., Murray-Clay R., Klahr H., **Baehr H.,** 2020, *Requirements for gravitational collapse in planetesimal formation — the impact of scales set by Kelvin-Helmholtz and nonlinear streaming instability***,** ApJ, 895, 91

Martin R., Zhu Z., Armitage P., Yang C.-C., **Baehr H.**, 2021, *Kozai–Lidov oscillations triggered by a tilt instability of detached circumplanetary discs*, MNRAS, 502, 4426

Zhu Z., Jiang Y.-F., **Baehr H.**, Youdin A., Armitage P., Martin R., *Global 3D Radiative Hydrodynamic Simulations of proto-Jupiter's Convective Envelope*, MNRAS, 508, 453

Klahr H., **Baehr H.,** Melon Fuksman J. D., *Thermal instabilities in accretion disks II: Numerical Experiments for the Goldreich-Schubert-Fricke Instability and the Convective Overstability in disks around young stars,* ApJ accepted

Rice K., **Baehr H.**, et al., 2025, *Dust density enhancements and the direct formation of planetary cores in gravitationally unstable discs,* MNRAS, 539, 3421

Carrera D., Davenport A., Simon J., **Baehr H.**, et al. *Turbulence Inhibits Planetesimal Formation in Class 0/I Disks Subject to Infall*, ApJ, 990, 39

Baehr S., **Baehr H.**, 2026, *Entropy and Diffusion characterize mutation accumulation and biological information loss*, arXiv: 2510.07265

Murray M., Hall C., **Baehr H.**, Terry J., 2026, *Dust morphology under changing dust mass ratios in protoplanetary discs*, MNRAS, 547, 1

REVIEW CHAPTERS/ARTICLES

Instabilities and Flow Structures in Protoplanetary Disks: Setting the Stage for Planetesimal Formation.
Hubert Klahr, Thomas Pfeil, **Hans Baehr**, Andreas Schreiber; Handbook of Exoplanets, eds. Hans J. Deeg and Juan Antonio Belmonte

Numerical Radiation Hydrodynamics for Circumstellar Disks: How to successfully set up and test your code for studies of planet formation
Hubert Klahr, **Hans Baehr**, David Melon Fuksman, Thomas Pfeil; Living Review of Computational Astrophysics

INVITED TALKS

Gravitational Instability in the Planet Formation Paradigm
University of Tübingen Astrophysicalisches Colloquium; Tübingen, Germany, November 14, 2016

Modeling Planet Formation in Young Self-Gravitating Disks
Formation and Dynamical Evolution of Exoplanets; Aspen, Colorado, USA; March 31, 2017

Self-Gravitating Disks and Planet Formation
2017 MIAPP: Protoplanetary Disks and Planet Formation and Evolution; Munich, Germany; May 29, 2017

Gravitational Instabilities: The Fastest Way to Form Planets
MPIA Königstuhl Colloquium; Heidelberg, Germany; June 14, 2019

Gravitational Instabilities and the Formation of Giant Planets on Wide Orbits
University of Warwick Seminar; Preston, UK; February 11, 2026

GRANTS/PROPOSALS

Proposal for the Extension of Computing Time with the Jülich Supercomputing Centre: JUQUEEN
Title: *Gravoturbulent Planetesimal Formation*
Principal Investigator: Dr. Hubert Klahr
First renewal: **46 million** cou-hours; May 1, 2016 – April 30, 2017
Second renewal: **36 million** cpu-hours; May 1, 2017 – April 30, 2018

Application for Computing Time with the Jülich Supercomputing Centre: JUWELS
Title: *Gravoturbulent Planetesimal Formation*
PI: Dr. Hubert Klahr
Period: **2.7 million** cpu-hours; May 1, 2018 – April 30, 2019

Contributor to ALMA Proposal *Witnessing Giant Planet Formation in the Act*
PI: Charles Law
Status: short baselines completed in ALMA Cycle 9; resubmitted for long baselines in ALMA Cycle 10

COMPUTER SKILLS

Word processing with Microsoft Office and LaTeX. Coding with Python, C++ and FORTRAN.

Several years of experience using and modifying hydrodynamic mesh codes including PENCIL, PLUTO and Athena++.

TEACHING EXPERIENCE

Substitute lecturer, *Astronomy 1010*, University of Georgia
September 2022 – August 2024

Teaching Assistant, *Practical Course in Numerical Methods*, Universität Heidelberg
February 8 – February 19, 2016

Assisted in the supervision of the following students:
Bhavya Joshi, June 2017 – October 2018
Master thesis on the implementation of new methods in an adaptive mesh code
Dominic Batzler, October 2017 – July 2018
Bachelor thesis on the behavior and evolution of solids in 2D local fragmentation simulations
Konstantin Gerbig, December 2018 – September 2019
Master thesis on planetesimal formation via direct gravitational collapse of particles

OBSERVATIONAL EXPERIENCE

17 nights remote transit photometry with EDEN project, 1.23m telescope at Calar Alto, 2019-2020 (See Gibbs et al. 2020 for details)

OTHER RESPONSIBILITIES

Refereed for MNRAS, ApJ, ApJL, A&A
Reviewer/Panelist for NASA XRP Grants and NASA Emerging Worlds Grants

RESEARCH INTERESTS

Planet and Star Formation
Circumstellar/Protoplanetary Disks
Self-Gravitating Systems
Hydrodynamic Simulations